

Rockefeller University
Welch Building – The Great Hall
1230 York Avenue, New York

ADVANCED GENE MAPPING COURSE

January 11-15, 2027

The course will be held every day from 9:00AM - 5:00 PM The Great Hall of the Welch Building on the Rockefeller University campus, located at 1230 York Avenue (entrance at 66th Street). On Monday evening a social gathering will be held at The Rockefeller University Faculty Club.

Each session will start with a theoretical introduction followed by practical exercises. Instructors for the course are Heather Cordell (Newcastle University), Andrew Dewan (Yale University), Suzanne Leal (Rockefeller University & Columbia University), Shamil Sunyaev (Harvard University), and Gao Wang (Columbia University).

A lecture will also be given on ethics and the regulation of human subject research. Additionally, a special guest lecture will be given by Jurg Ott (Rockefeller University).

MONDAY – January 11th

Instructors – Heather Cordell (Morning) and Suzanne Leal (Afternoon)

Statistical framework for association testing common and rare variants (population and family based); linear and logistic regression; variance components analysis; linear mixed models (LMM) and generalized LMM; Estimating heritability and genetic correlations; Controlling for population admixture and substructure; Data quality control of genotype and sequence data; Annotation of sequence data

Exercises: PLINK, GCTA-GREML, LD Score Regression, FaST-LMM

Special Lecture

Jurg Ott, Ph.D. (1:00-2:00)

Odds ratio-driven polygenic risk scores

Evening Program

5:15 PM Get-together (light food and beverages) for course participants and instructors at The Rockefeller Faculty Club

TUESDAY – January 12th

Instructors - Heather Cordell (Morning) and Suzanne Leal (Afternoon)

Rare variant association analysis using exome and sequence data for population- and family-based data; rare variant association tests; meta-analysis, detecting gene x gene and gene x environment interactions, Imputation of genotype data and its analysis, power and sample size estimation.

Exercises: REGENIE, Genetic Power Calculator, R, and Cassi

Special Lecture - Responsible Conduct of Research

Instructor - TBN (4:00-5:00)

Conflict of interest, research ethics, protection of human subjects, and data security

WEDNESDAY – January 13th

Instructor – Gao Wang

Molecular QTL mapping; Transcriptome-wide association studies (TWAS)

Exercises: TWAS

Statistical fine mapping (concepts and application: summary statistics, functional annotation, colocalization and multi-trait analysis).

Exercises: SuSiE and Multivariate fine-mapping (mvSuSiE)

THURSDAY – January 14th

Instructor – Andrew DeWan (9:00-5:00)

Biological, mediated, and spurious pleiotropy; Methods to detect pleiotropy; Mediation analysis; Mendelian Randomization

Exercises: PLINK2.0, MultiPhen, Mediation, MR-PRESSO, TwoSampleMR

FRIDAY – January 15th

Instructor – Shamil Sunyaev (9:00-5:00)

Prediction of variant functionality; analysis of epigenomic data; estimation of polygenic risk scores